

Multifunction RECHARGEABLE DIGITAL CLOCK

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Presented here is a digital clock that also functions as a rechargeable light lantern and a mobile charger. The clock has all the basic functions like date-time display, alarm and room temperature display. It is powered by a lead-acid battery, which can be recharged through solar power or mains power supply. So the clock never runs out of power.

The clock is provided with a female USB socket, which can be used to charge most of your mobile phones and gadgets. Its dedicated battery charging and load cut-off circuitry takes care of the battery through exact charging topology and prevents it from getting over-charged or under-charged.

Block diagrams

The functional unit of the clock is divided into power unit and main clock unit. Block diagram of the power unit is shown in Fig. 1.

The power unit provides continuous power to the clock and other circuits. Its lead-acid battery can be charged through solar power or AC mains power. The battery management circuit is designed to accept power from two sources at the same time but gives priority to the mains power. So if the solar power is low or unavailable, users can plug the battery into the AC mains wall supply to charge it.

A solar photovoltaic panel is used to directly convert solar light and heat energy into electricity. To harness AC mains power for the circuit, a step-down transformer is used. It converts high-voltage AC into low-voltage AC. This AC voltage is rectified and

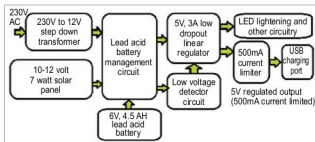


Fig. 1: Block diagram of power unit

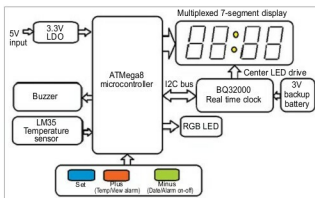


Fig. 2: Block diagram of main unit

converted into DC, and then further smoothed by using capacitors as low-pass filters. After this, the power becomes acceptable for the battery management circuit.

The lead-acid battery is directly connected to the battery management circuit, which takes care of battery charging with charging voltage and current maintained to acceptable limits. This ensures longer battery life, capacity and optimal usage.

Next, a comparator circuit is used. It checks the battery voltage against its internal reference voltage and shuts off the load, thus preventing the battery from further discharging. A linear low-dropout (LDO) voltage regulator is used to derive regulated 5V from the battery rail. This 5V rail is non-current-limited and powers most of the circuitry ahead, including LEDs. A

current-limited load switch is used to derive 5V rail, which is current-limited to 500mA for USB charging applications in order to prevent USB application from immediately discharging the clock battery.

Block diagram of the main clock unit is shown in Fig. 2. The microcontroller in the circuit keeps track of all the sensors and switches connected to it, makes decisions and calculations, and finally updates the display section. In this section, the non-current-limited 5V rail from the power section is converted into 3.3V rail via a 3.3V LDO voltage regulator (UCC383) that powers the microcontroller and other peripherals.

To sense the room temperature, a temperature sensor is used. Its output is an analogue signal proportional to the temperature. This signal is fed to one of the analogue-to-digital converter channels of the microcontroller. The user inputs are sensed by three momentary push-to-on switches, which are used to set/check various parameters of the clock like alarm, date, etc.

Time keeping is done by a separate real-time clock (RTC) chip that is interfaced to the microcontroller via I2C bus. The RTC is also powered by a backup battery, which keeps the time running even if the AC mains power fails.

The display section comprises four 7-segment displays, which are multiplexed to efficiently use the limited pins available on the microcontroller. The two centre LEDs between the two pairs of 7-segment displays are driven by the RTC clock. An RGB LED is used to display AM/PM and date. Finally, a